Investigation for re-purposing DART Network to study the Thermohaline Circulation

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Abstract

Understanding the deep ocean is critical for studying a changing world, however deep ocean data remain spatially and historically limited. The GeoNet's Dart Network is an early warning detection system near New Zealand. The bottom pressure recorder has a temperature sensor placed inside the device, recording internal temperature. The objective of this report is to investigate the DART Network temperature data and determine if it can be used as direct water temperature measurements. Anomalies and patterns in the temperature data were investigated across different sensors and timescales, as well as compared to reliable Argo sources. The result showed that there was a lot of noise, but also useful areas in the data. A sinusoidal pattern was captured in the diurnal cycle, and a similar temperature range to Argo data suggests that the DART network is able to capture changes in temperature, despite having an incorrect direct base temperature measurement. The data is less reliable when used below an hourly timescale, as the software seems to have some type of hourly process that consistently influences the temperature data on the hourly scale.

Acknowledgements

I would like to acknowledge Melissa Bowen for supervising and providing the initial idea for this research project.

Introduction

The Thermohaline circulation is how the ocean distributes heat, salt, nutrients and marine life all around the world. Understanding the Thermohaline Circulation is crucial for predicting climate patterns, sustaining marine ecosystems, and monitoring global heat transport. Despite its significance, data on deep ocean currents remains limited due to the vast and deep nature of the ocean.

Advancements in technology have led to greater data available through the Deep Argo Program, deep mooring techniques, and occasional ship casting and survey data. However, coverage and historical data of the ocean floor remain limited. An existing deep ocean monitoring system, the “Deep-ocean Assessment and Reporting of Tsunami” (DART) Network, has shown some potential in improving deep ocean visibility.

The DART Network is an early Tsunami detection system in New Zealand; it consists of a Bottom Pressure Recorder (BPR) on the ocean floor to measure changes in water pressure. Within the BPR is a temperature sensor used to monitor internal component temperatures. This research project investigates the DART Network internal temperature sensor data and its viability to be re-purposed to be used as a direct measurements for local deep water temperatures. If possible, it will improve the historical temperature data available and increase the data coverage along the ocean floor.

Literature Review

The Thermohaline Circulation, often referred to as the Global Conveyor Belt, is a series of large, interconnected ocean currents that control the movement of the world’s ocean water. Warm surface waters are transported into deep underwater currents along the ocean floor, some of these currents takes thousands of years to resurface. The Thermohaline circulation spans across the 5 oceans, redistributing heat, salt, nutrients, dissolved gases, and marine life around the world. (Pauthenet et al., 2019).

Thermohaline Circulation Structure

The circulation is mainly driven by the differences in water density, which is a function of the water's temperature (thermo) and salinity (haline). Water density is lowest towards polar regions and the bottom of the ocean, where temperatures are typically low and salinity is high. Although this is not always the case, such as the subtopics having the saltiest surface water, despite the polar water still being denser. This is due to temperature having a stronger influence on density compared to salinity. (Zika et al., 2015).

The structure of the Thermohaline Circulation can be understood through the "push" and "pull" theory that drive the circulation, described by Lozier (2010). The theory is a way to categorise the driving forces of the Thermohaline Circulation. The "push" theory suggests that the circulation is pushed by deep water formation in the polar regions, as new deep water is created, it pushes existing deep water through the deep circulation currents. Deep water forms during sea ice creation, when water freezes, only pure water is frozen out of the ocean. The salt content is left behind, making the remaining local water saltier. Combined with the low temperatures of the polar regions, the water becomes dense and sinks to the bottom of the ocean.

When deep water forms in the North Atlantic, it is known as the North Atlantic Deep Water (NADW), which sinks and travels south along the western coasts of North America and South America, as shown in Figure 1. Deep water often travels along the western side of continents due to the Coriolis Force; scientists have referred to these currents as the Deep Western Boundary Currents (DWBC). It will continue along this DWBC until it reaches the Antarctic Circumpolar Circulation (ACC). The Antarctic has a similar deep-water formation during it's own sea ice creation, the deep water sinks forming the Antarctic Bottom Water (AABW), which also connects to the ACC.

ACC given by the name, is a current that circumnavigates the Earth surrounding the Antarctic. It is a unique current in that it has both a surface and a deep current that flow in the same direction. Due to the lack of landmass, there are only a few places where the ACC deviates, one is South of Africa, and the other is along New Zealand. Bottom water leaves the lower limb of the ACC in these locations and creates the DWBC along the African continent and the Pacific Islands.

Along these two DWBC, upwelling occurs, causing the deep water in the lower limb to rise to the surface. The "pull" theory here suggests that part of the

Thermohaline is driven by the pulling force from these up-welling forces. There are many factors for upwelling, common ones are the wind-driven Ekman transport, warmer latitude, and bathymetry.

After the deep water has upwelled, it now travels eastward along the surface currents, driven by the prevailing winds in the low latitudes. It will continue until it reaches the Atlantic Ocean, at which point it will head North to fill in the surface water that had been used in deep water formation in the North Atlantic and continue the cycle.

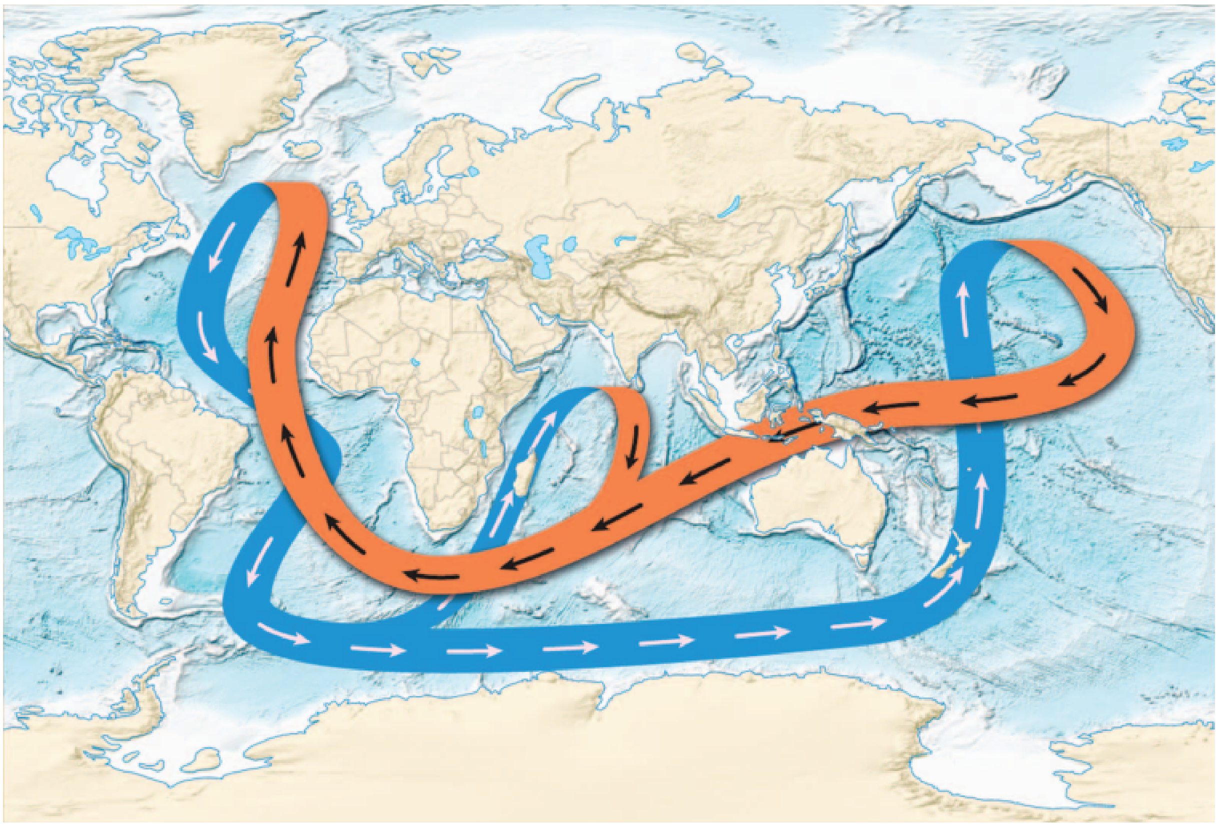

Figure 1. Diagram of Thermohaline Circulation. Orange represents warm surface water, while blue represents cold deep ocean water. [Adapted from Lozier (2010).

Thermohaline Circulation around New Zealand

Temperature and salinity are the main driving forces of the Thermohaline. However, there are many other influences on the Thermohaline circulation, such as bathymetry, wind currents, climate modes, Coriolis force, landmass, and numerous other global and local factors that affect specific currents.

This research paper will focus on the Thermohaline Circulation around New Zealand. New Zealand is situated near the strongest ocean current on Earth, the Antarctic Circumpolar Current (ACC), and serves as the entry point to a Deep Western Boundary Current (DWBC). The ACC is a current that circles the entire Antarctic continent. Unlike other currents, the ACC encompasses the entire vertical water profile, consisting of both a warmer surface current and a cooler deep current that flows eastward (NIWA, 2008).

On the east side of New Zealand sits another important current of the Thermohaline Circulation, the Deep Western Boundary Current (DWBC). This current takes Antarctic Bottom Water (AABW) from the lower limb of the ACC around the east side of NZ and into the DWBC in the Pacific Ocean, where it will resurface during upwelling (NIWA, 2008).

When sea ice forms at the coast of Antarctica, only pure water is frozen while the salt is rejected. This salt sinks into the deep ocean, increasing the salinity and density of the deep waters. Katabatic winds from the Antarctic Mountains, comes down and blows away newly formed sea ice, creating a continuous cycle. This cycle of increasing salinity and decreasing temperature causes the coastal water to sink due to density and travel northwards, providing the dense water for the AABW (Silvano et al., 2023).

The reason the ACC deviates around New Zealand is due to it southern position but also the local bathymetry. This mainly refers to the wider submerged continent New Zealand sits upon, Zealandia, distinguished by its higher elevation above the surrounding oceanic crust (Mortimer et al., 2017).

As a sunken continent, its elevation adds to the bathymetry, influencing surrounding currents. South of New Zealand is the Macquarie Ridge, an underwater mountain ridge that is an extension of the on-land Southern Alps. The ridge sits along the ACC path, disrupting it and serving as the entry point for AABW into the DWBC. (NIWA, 2008).

East the New Zealand is the Campbell Plateau and the Chatham Rise. These are elevated pieces of underwater landmass extending eastward, outlining the sunken Zealandia continent. After the AABW passes through the Macquarie Ridge, the bathymetry deflects it northeast toward Campbell Plateau. Due to the Campbell Plateau's elevated height, the AABW flow along the edge, concentrating and intensifying the flow. This continues north along the Chatham Rise, reaching the eastern edge where it will enter the Pacific Ocean, providing the dense water in the DWBC (Carter, 1997).

Data Availability

These areas play a crucial role in the heat, salt, nutrient, and biological processes surrounding New Zealand, and are worth researching and measuring. Data on the Thermohaline Circulation is abundant for certain areas, while it is extremely scarce in others. Technology such as satellites enables surface current data to be abundant, but limited accessibility for the deep current on the ocean floor (Carter, 1997).

This research report will be about data near New Zealand, and as such, we will look at the data availability near. The most abundant data available, including around the world, is sea surface temperature data from satellites. This type of data collection is extremely efficient and has a large area of coverage. Radiation emitted from an object is dependent on its heat content, as long as the satellite has a line of sight of the ocean, it is able to measure it temperature. This type of data is collected and openly available from many sources, such as NIWA (n.d.-b) or MetService (n.d.).

Another type of data available around New Zealand is from Argo. Argo are devices at sea that float at various depths up to 2000m below surface level. These devices can record various information, including temperature and salinity, at different layers before resurfacing and transmitting this information to satellites. Argo data is part of the international Argo Program that standardises and opens source all the information collected around the world. There are around 3,900 active Argo, and it is by far the greatest source for below-surface data. The New York Times have described the Argo Program as “one of the scientific triumphs of the age”. But even Argo has their limitations, the typical Argo depth is limited to 2000m, and the position of the Argo is dependent on the currents and cannot take continuous stationary samples. There is a Deep Argo Program that can reach depths of 6000m; however, currently there are only ~200 active Deep Argo and even few near New Zealand (Argo Program, n.d.).

Mooring is another data collection technique that has been used around New Zealand. This involves attaching measurement devices to a chain connected to an anchor on one end and a buoy on the other. The device is left at stationary point at sea for a period of time, collecting a vertical information profile. This has been done at the Macquarie Ridge, Campbell Plateau, and Kermadec Trough near New Zealand. The limitation of Mooring is the small area of coverage, the fixed data period, and the extremely high financial & labour cost. Additionally, many places are too dangerous to use mooring techniques, such as the ACC due to the strong waves and stormy weather (NIWA, 2008).

Objectives and Methods

GeoNet is a New Zealand's national geohazard monitoring system, one of their systems is the DART network designed to monitor tsunamis. It does this through a device the bottom of the ocean floor that detects sudden changes in pressure, additionally it also has a internal temperature recorder. My research supervisor, Melissa Bowen (2025), was first to notice that the temperature data was mislabelled, it was listed as water temperature when the sensor was placed inside the housing unit, and did not have direct contact with water. The objective of my research project is to investigate this internal temperature dataset and assess the viability of pre-purposing it as a measurement of surrounding deep water temperatures.

DART Network

The Deep-ocean Assessment and Reporting of Tsunami (DART) Network uses devices on the sea floor to detect Tsunami as a part of an early warning system. The device on the sea floor is called the Bottom Pressure Recorder (BPR), which uses the change in water pressure to detect incoming Tsunamis, typically intended for use after natural events such as Earthquakes. When a change is recorded, the BPR uses an acoustic signal to communicate with a buoy, which in turn uses a faster electromagnetic signal to communicate with satellites, which transfers all this data to an operating centre. (Geonet, n.d.).

The BPR has two modes. In its normal mode, it records data samples in 15mins intervals; the samples are packaged and transmitted every 6 hours. Once a significant change in water pressure is detected, it switches to event modes. In event mode, data is sampled every 15 seconds and is immediately transmitted; this continues for the next 3 hours before switching back to normal mode. It should also be noted that it does record temperature data but is not transmitted; it is stored locally and is only available after maintenance, which occurs once every two years.

Objectives

My research objective is to investigate the viability of using the DART Network's BPR internal temperature data as a measurement for the temperature of the surrounding deep water. The research objects were:

1. Investigate the temperature pattern during Event Mode
2. Investigate the temperature spike at the beginning of the data to see if a vertical temperature profile be created
3. Investigate the differences in temperature between sensors
4. Investigate the data reliability by comparing it to Argo data
5. Investigate any natural and human patterns in the data

Methods

Most of the investigation will be conducted using one sensor at one station for the sake of time and efficiency. Since I am motivated to investigate the dataset to study the Thermohaline, the station closest to it was chosen. There are five stations close to New Zealand; however, all 5 of these are north of the Chatham Rise and are not along the east edge where the AABW travels. Station NZE was chosen because it is the easternmost of the five and will most likely be the closest to the DWBC as it travels towards the Pacific Ocean.

1. The data used in this investigation is from GNS Science (2023) as part of the New Zealand GeoNet program sponsored by NHC, LINZ, NEMA and MBIE. I used the GeoNet API to access the data, while using the Python packages of pandas and matplotlib to graph and do correlation analysis on the three main datasets of temperature, pressure, and height.
2. To investigate the possibility of creating a vertical temperature profile with the data during descent, I will graph the temperature and height during the descent period.
3. To investigate the differences in data between sensors, I created a function to simultaneously graph all the available sensors within a station with each other.
4. To investigate the reliability of the data, I will compare it to deep Argo data, a well-known, reliable source for deep ocean data (Argo, 2025). To access this data, I will use the Python package argopy.
5. To investigate patterns, I will analysis the data on time-scales with known natural patterns, such as diurnal, lunar, and seasonal timescales.

Results and discussion

Plotting temperature data during both normal and event mode, reveals anomalies that will be investigated. Figure 2 shows normal and event mode temperature data plotted together. There are two types of spikes I was interested in, one is the spikes associated with the event mode, and the other is the spike at the start, possibly related to the deployment. Additionally, there is a spike in normal mode that didn't correlate with a similar event mode response, around August 2021, which will also be investigated.

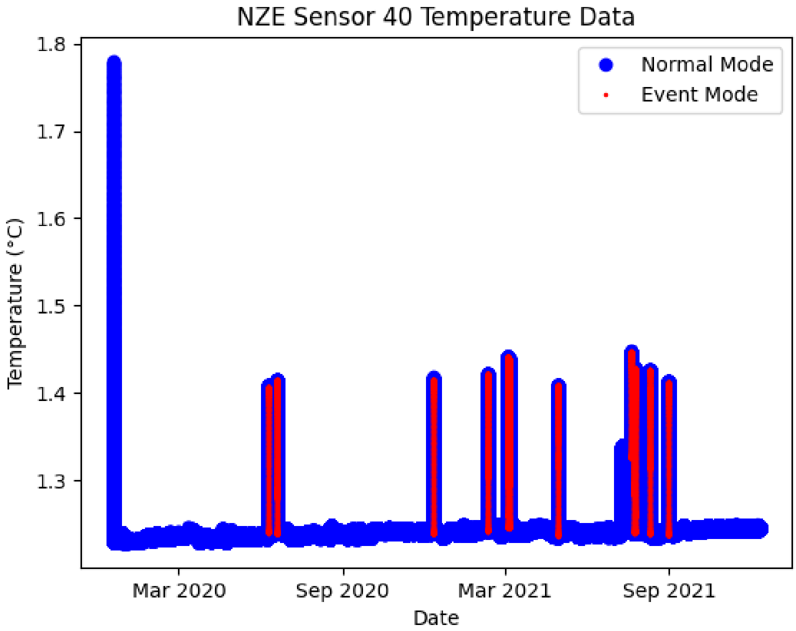

Figure 2 Normal mode temperature data in blue and event mode temperature in red

When event mode is triggered, there is a consistent, sharp increase in temperature. The temperature pattern at which the event mode ends is not as consistent; however, it is always after the peak. This shows that the switch to event mode does cause the device to produce more heat, affecting the data. If this data is to be used to study the deep ocean, data relating to event-mode should be cleaned. It should be noted that the temperature does not immediately return to normal mode upon event mode termination, as shown in Figure 3. This means that simply excluding data associated with the exact time event mode starts and ends will still leave traces in the data. It should be additionally noted that event modes have been triggered without a sharp change in pressure; this could be done for testing purposes, regardless the data is categorised under event mode.

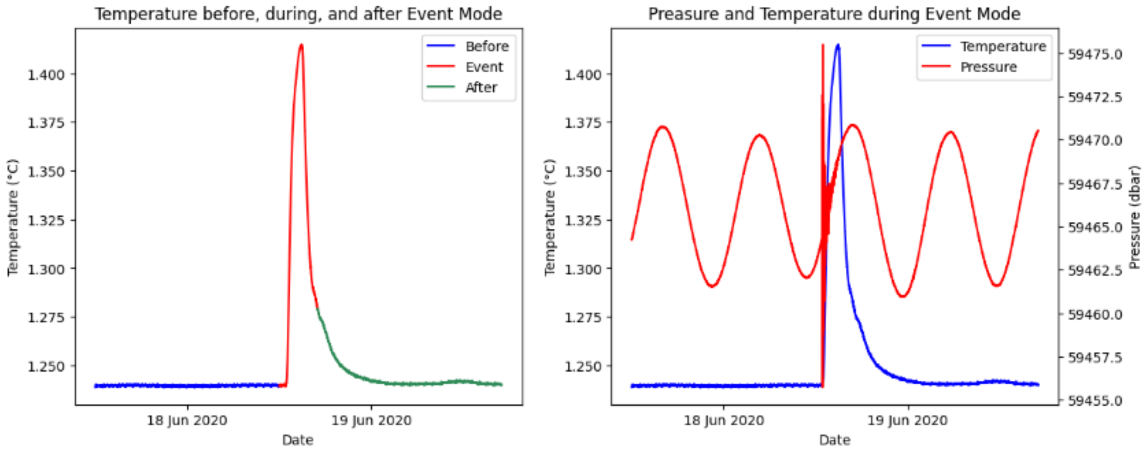

Figure 3 – Left highlights the period before, during, and after the event mode. The right shows the corresponding changes in pressure during the period

Investigating the temperature spike at the beginning of the data was to see if the BPR had recorded temperature data during it's descent. The hope was that if it did, height data might have also been recorded, the combination of height and temperature data would means that a vertical temperature profile could be formed.

Figure 4 shows the first 7 days of temperature and height data revealing a negligible difference in height throughout the 7 days. Though height peaked at the start, the range of height was less than 2m, nowhere near the amount needed for this degree of change in temperature. This temperature spike is likely related to the testing or set-up process of the device, rather than a temperature difference due to different layers of the ocean.

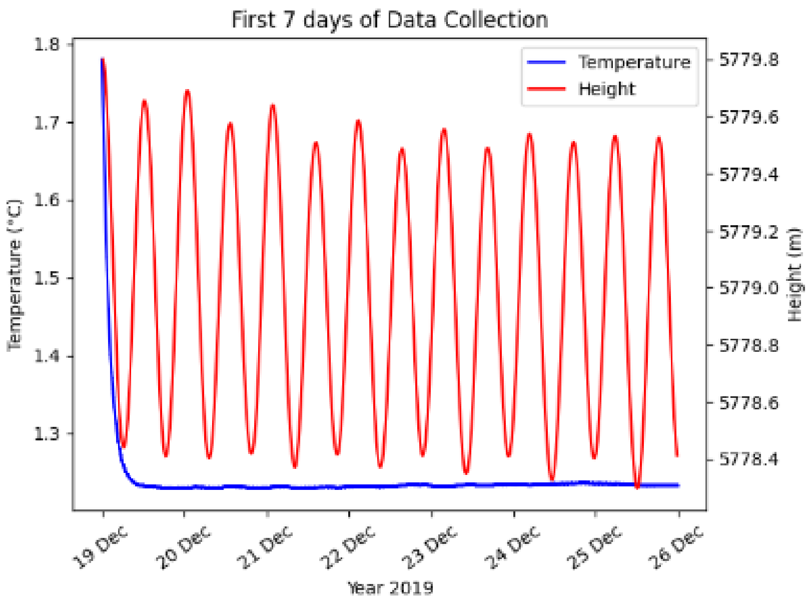

Figure 4 – The first 7 days of data, temperature on the left y-axis and height on the right y-axis

On July 9, 2021, a weird anomaly appeared in the data, the temperature increased and remained between the range of 1.30°C – 1.35°C for 13 days, see figure 5. After the 13 days, an event mode was triggered without a change in pressure. After the event mode ended, it returned to its regular temperature. This correlates to the increase in temperature in Figure 2 without an event mode correlating with its duration. This is important to mention as it is another type of anomaly that would need to be investigated if the dataset were to be used. From the nature of the pattern, this could have been a software update that triggered the temperature increase and ended with a test of the event mode. Alternatively, something could have been wrong with the device, and a manual trigger of the event mode fixed it.

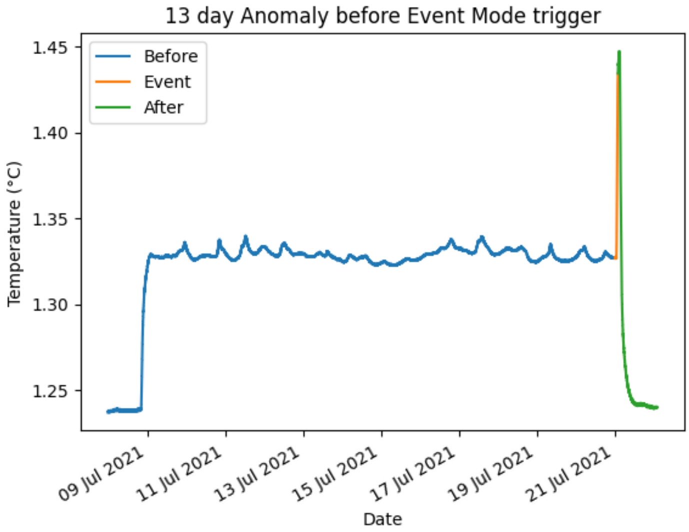

Figure 5 – Mysterious increase in temperature on 9 July 2021

A correlation graph between temperature and pressure yields interesting results in Figure 6. The most eye-catching feature on the graph is the line that forms toward the top right. This line correlates to the spike at the start of the data. The peak pressure along this line is also similar to other points in the graph, which further supports that the initial temperature spike is not due to temperature recording of different layers of the ocean.

The next most interesting feature on the graph is the box-shaped pattern. The top and bottom walls of the box are determined by pressure. The position of these walls likely relates to the natural peaks and troughs of waves. The left and right wall of the box is determined by temperature. The left wall is likely the normal state for the BPR, given how many points resides along it. The right wall is likely the peak temperature during an event mode trigger; the horizontal parabolic patterns with similar temperature maximums also supports this idea.

Other noticeable patterns are the large number of points on the left wall that are high and low in pressure. These points would have had a steady increase or decrease in pressure such that there was no sudden change in pressure to trigger event mode, allowing it to be recorded with low temperatures. The last noticeable pattern is in the middle of the box. There is a large number of points along a single temperature value between 1.3°C – 1.4°C. This likely correlates to the previously mentioned anomaly on July 09 2021, as it is the only consistent data point at this temperature value.

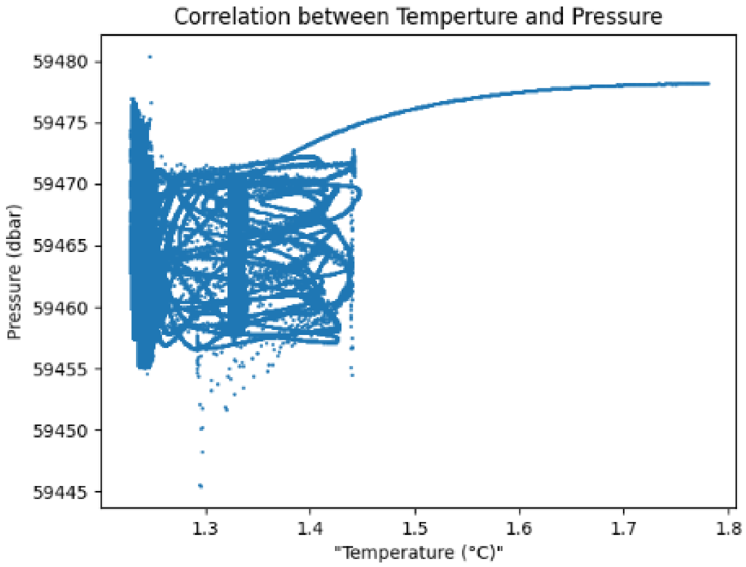

Figure 6 - Correlation graph between temperature and pressure

Pressure is a function of depth in the ocean; the deeper the depth, the greater the pressure. As expected, examining the correlation between pressure and height data reveals a strong correlation.

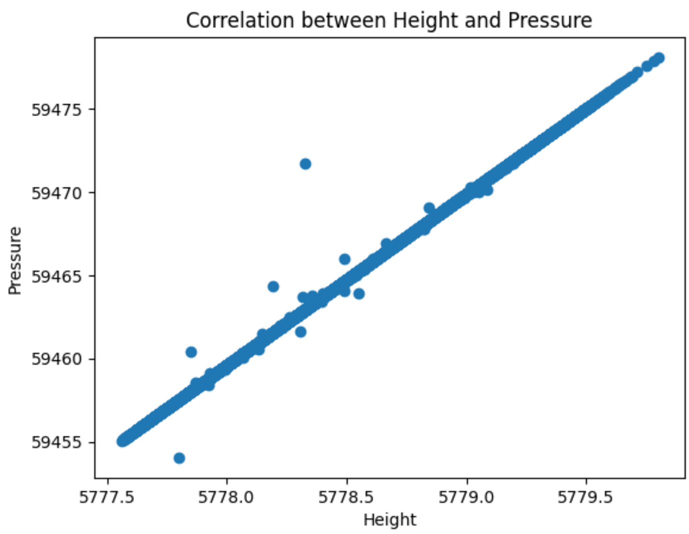

Figure 3 - Correlation graph between height and pressure

Sensor Comparison

Now that we understand the data of station NZE's sensor 40, we can start comparing it with other sensors. The NZE station has 3 sensors, 40, 41, and 42. Sensor temperature data is stored locally on the BPR and is only available after resurfacing during maintenance. As of this report, station NZE is active with sensor 42. This means that only temperature data for sensors 40 and 41 is available, while sensor 42 is not yet available. Comparing the temperature data of these two sensors in Figure 8, clear differences are evident. The average of sensor 41 is 0.10°C higher than sensor 40.

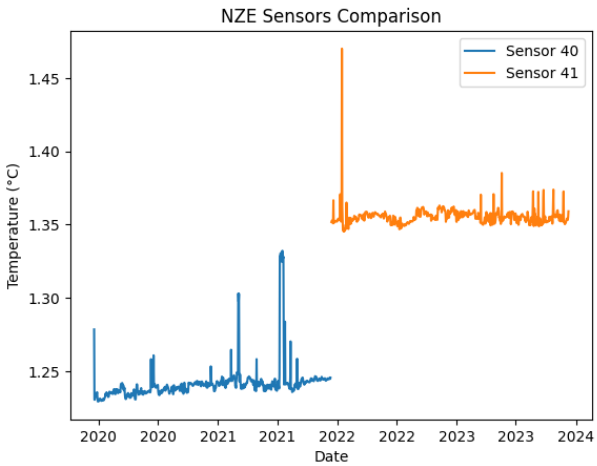

Figure 8 – Daily temperature data of station NZE sensors

	NZE 40	NZE 41
Latitude	-36.0493°	-36.0399°
Longitude	-177.708°	-177.697°
Height	-5779m	-5777m

Table 1 – Location of station NZE sensors

Using the coordinates of each sensor provided by GeoNet, see Table 1, a simple Pythagoras calculation can be done.

$$\sqrt{(111.320\text{km} \times 0.0009°)^2 + (111.320\text{km} \times \cos(0.0009°) \times 0.0011°)^2} = 0.158\,km$$

Sensor 40 and sensor 41 were placed 0.158km apart with sensor 41 being elevated 2m higher. Considering the vastness of the ocean and the stability of deep ocean temperatures, it is highly unlikely that a 0.158km displacement would cause a nearly ~10% change in temperature. It is more likely that some change in the equipment, be it the main components or the temperature sensor itself.

Figure 9 examines the temperature data across all sensors from every station, a shift in temperature occurs after every maintenance. Some stations experienced similar increases in temperatures, others experienced a decrease in temperatures, some looked to have no significant change, while station NZJ nearly had a 2°C jump in temperature. All the maintenance seems to have been done at three distinct points in time, one near the start of 2022, one in the middle of 2022, and one in the middle of 2023. All of which resulted in varying changes in temperature, and each set of maintenance showed no trends. It is difficult to say what the cause is just from the data alone. If I had to guess, it would be that the temperature sensor got shifted, assuming there is no standard location for it. It is unlikely the cause being equipment upgrades, as it would likely show some type of trend. If this data were to be used as accurate deep ocean temperature measures, a lot of work would need to be done to recalibrate each sensor's data to an accurate norm.

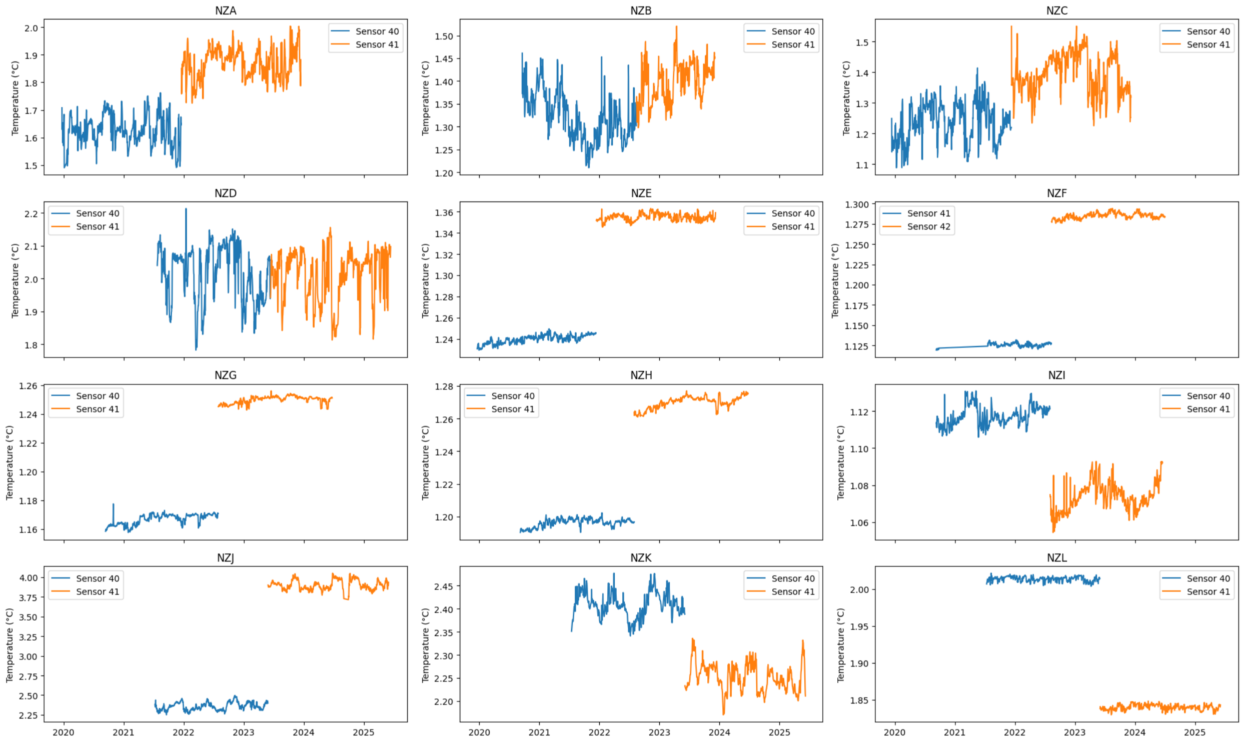

Figure 9 – Comparison of sensor at each station with outliers removed

Data reliability & Argo Data

Now that we understand what is going on within the datasets, we can examine their reliability in comparison to other, more well-known data sources. From the deep Argo data, there are some data points that are close to the location of our station that can be used. Comparing the available data points to the GeoNet data on Figure 10, we can see a clear disparity in the temperature profiles. The GeoNet data is consistently ~1°C higher than the Argo data. This outcome is not surprising, considering that sensors have varying temperature readings; it would not be reasonable to expect one to accurately record the true temperature unless it were by chance.

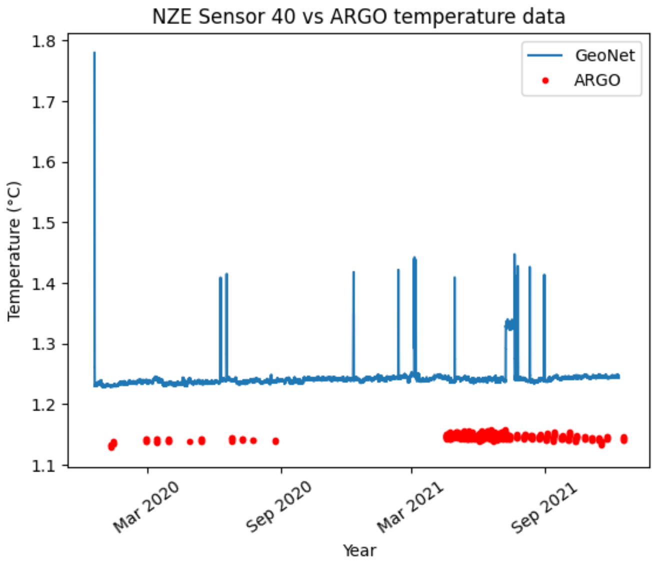

Figure 10 – Argo temperature data in red and DART temperature data in blue

Though direct temperature measurements are not reliable, changes within the data could be useful. Take the range shown in Figure 11 for example. Argo had a data range of 0.028°C, while NZE sensor 40 had a data range of 0.025°C. The data ranges are quite similar; this could mean that changes in temperature data could tell a story about the surrounding temperatures. Having a similar range is not enough to make it reliable for representing temperature changes; it is uncertain whether this is caused by changes in the surrounding water temperature or if the device simply produces heat, causing a similar range.

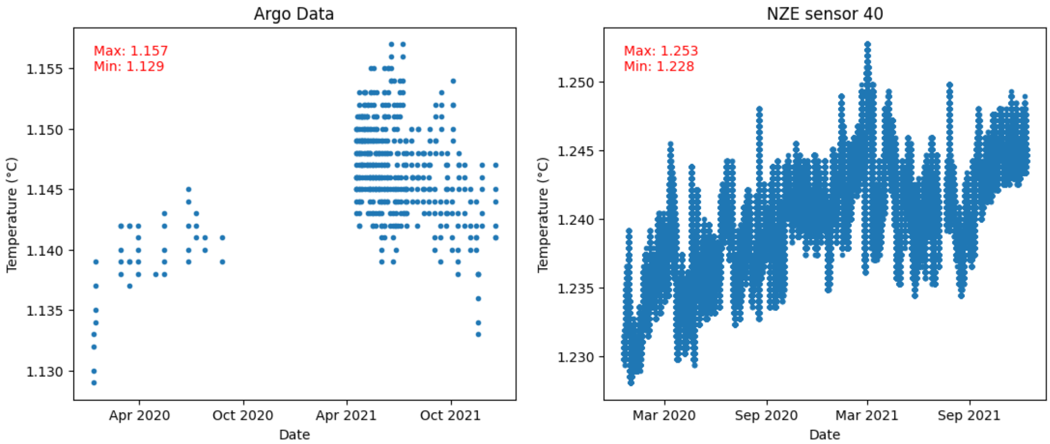

Figure 11 – Left is the Argo data, and right is the DART data with outliers removed. The maximum and minimum range is the top right of each graph

Data Patterns

Examining diurnal variation reveals clear natural and human patterns. Each hour starts with a temperature spike and ends with a temperature drop. This pattern is consistence every hour without fail, and is highly likely to be related to the operation of the device. Without more details of the inner workings of the device from GeoNet, it is difficult to determine if there are any unaffected data points within each hour. As previously mentioned, the DART network transmits data every 6 hours in normal mode (Geonet, n.d.). Interestingly, I could not find any 6-hour pattern with temperatures. Perhaps the online information is incorrect, or the transmission of data produces very little heat.

The natural pattern within a diurnal cycle is a sinusoidal pattern. This pattern is present each day, but the strength and definition varies. The sinusoidal shape leads me to believe that this is caused by the Earth's spin and the gravitational pull of the Moon and Sun, similar to the tidal bulges. Further investigation with lunar and solar phase data could yield very interesting results.

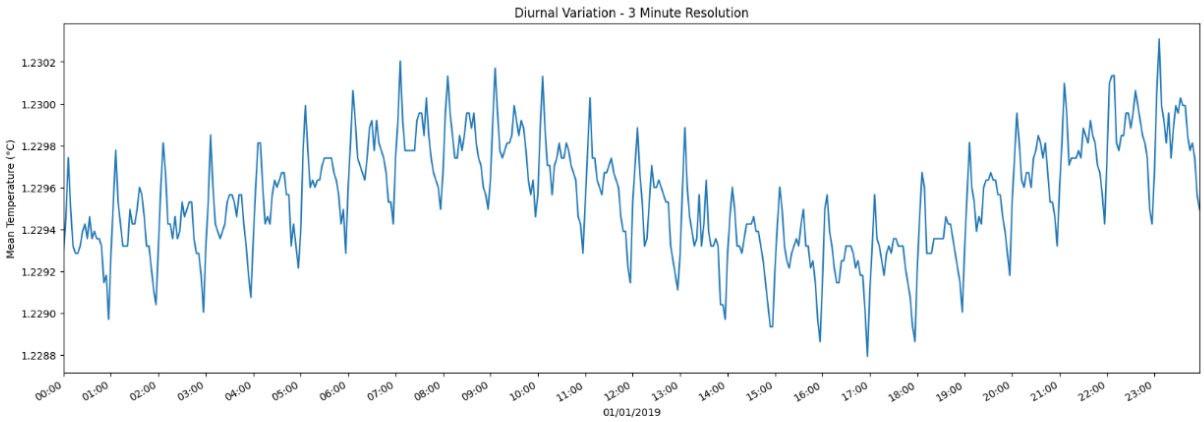

Figure 12 – The daily temperature on 01/01/2019 with points grouped into 3-minutes intervals

Examining the data on a yearly time-scale, I found no discernible patterns. I specifically searched for yearly seasonal variations and monthly lunar variations. Considering the gravitational pull theory suggested when discussing the diurnal variation. The lack of patterns could suggest that the moon and tilt of the Earth have little influence compared to the sun for deep ocean temperatures.

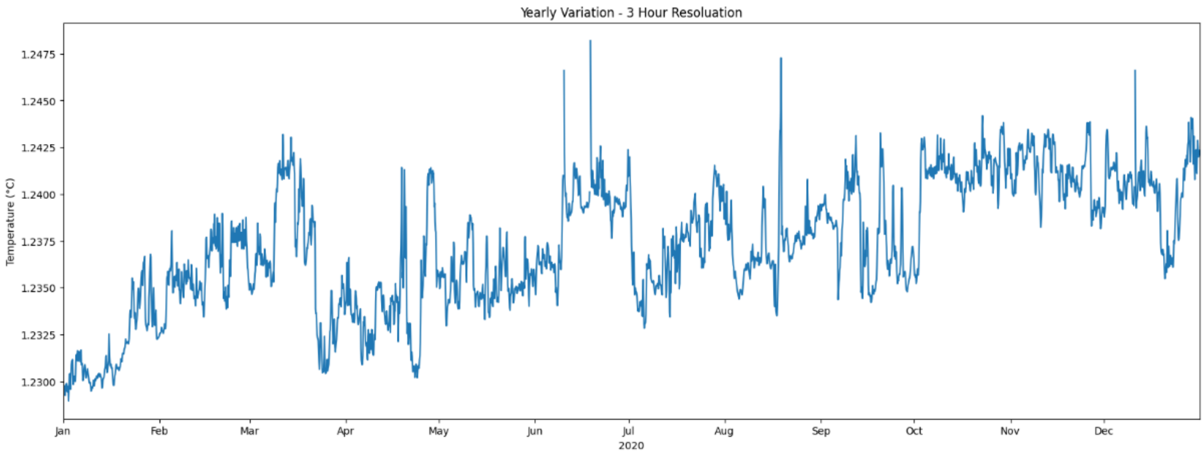

Figure 13 – 2020 yearly temperature with points grouped into 3-hours intervals

Conclusion

The DART Network temperature data from GeoNet contains a significant amount of noise and requires extensive processing to be useful. Such as the triggering of event mode, causing the temperature to rise. The temperature spike at the beginning of the datasets is caused by the initiation after deployment, and not temperature recording during the descension. Additionally, there is an unexplained anomaly on station NZE sensor 40 on the 9th July 2021. Using the data below an hourly resolution would need further work in understanding the inner software of the device.

The similar data range to the deep Argo data gives prospects for the dataset. A similar data range could indicate that the dataset can be used to detect temperature changes despite having a different direct measurements; further work is needed to understand the origin of the temperature in the device. The sinusoidal pattern in the diurnal cycle is evidence that there is useful natural patterns captured within the data. Further work with lunar and solar cycle data could further confirm this, this work could also be helpful in identifying if there are factors influencing the varying strength in the sinusoidal pattern.

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